

Lifelong Opportunity Guide 93 — Surveying and Mapping Technician

Version: 2.0 controlled working master

Language: English

Primary U.S. benchmark: O*NET-SOC 17-3031.00 — Surveying and Mapping Technicians

Canada comparison: NOC 22213 — Land survey technologists and technicians

Colombia comparison: CUOC 31122 — Técnicos en topografía

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What this career is

Surveying and Mapping Technicians help measure, document and represent the physical world. Depending on the employer and project, they may collect field measurements, operate surveying instruments, establish or check reference points, support construction staking, calculate angles/elevations/distances, process survey data, prepare drawings or maps, maintain records, work with GIS/CAD systems, interpret aerial or satellite imagery, and support licensed surveyors, engineers, cartographers or photogrammetrists.

The work sits between field measurement, mathematics, geospatial technology and technical documentation. Titles can include Survey Technician, Mapping Technician, Geomatics Technician, Instrument Person, Survey Assistant, Photogrammetric Technician, Topographic Technician and related titles.

A title alone does not define authority. A technician may perform important measurements and calculations without having the legal authority to establish, certify, sign or seal a property boundary. Always check the actual duties and the rules of the jurisdiction.

What technicians actually do

Current U.S. O*NET/BLS evidence supports work such as:

- collecting field survey data;
- measuring angles, distances and elevations;
- operating surveying/mapping equipment within assigned scope;
- setting, holding or checking targets, rods, prisms and reference equipment;
- calculating geographic positions from survey data;
- checking mathematical calculations;

- supporting construction staking/layout;
- compiling information from engineering plans;
- producing or updating maps;
- checking map layers for errors and completeness;
- creating or maintaining geographic/topographic databases;
- processing aerial, satellite or photogrammetric source material;
- documenting field observations and technical details;
- supporting engineers, surveyors, cartographers and other professionals.

Field and office work often connect directly: poor field notes or incorrect setup can create downstream errors that no CAD/GIS software can repair automatically.

Technician versus licensed or authorized surveyor

This distinction is essential.

Technician scope can include

- collecting measurements;
- setting up and operating equipment under assigned procedures;
- maintaining field notes;
- calculating coordinates, elevations and distances;
- processing survey observations;
- drafting or mapping;
- construction staking support;
- quality checks;
- data conversion;
- photogrammetric or drone-data processing where authorized;
- GIS/CAD support.

Licensed/authorized professional scope may include

Depending on the jurisdiction:

- legally determining property boundaries;
- certifying boundary positions;
- signing or sealing survey documents;
- issuing authoritative cadastral/legal opinions;
- supervising work where law requires professional oversight;
- representing survey conclusions as legally controlling.

BLS states that surveying and mapping technicians in the United States do not require surveying licensure, while experienced technicians with additional formal training may later qualify to become licensed surveyors.

Do not represent a technician calculation, GNSS coordinate, drone orthomosaic, CAD drawing or GIS layer as a legally authoritative boundary unless the required professional authority and evidence exist.

Measurement foundations

Survey work depends on disciplined measurement rather than simply reading numbers from an instrument.

Understand:

- distance;
- horizontal angle;
- vertical angle;
- elevation;
- slope;
- coordinate;
- bearing/azimuth;
- station/chainage concepts where used;
- benchmark/control point;
- traverse;
- backsight/foresight concepts;
- closure/error concepts;
- precision versus accuracy;
- tolerance;
- significant figures and units.

A measurement can be precise but wrong. Repeatability does not prove that the setup, control, calibration, coordinate system or reference point is correct.

Units and conversions

Survey work may use metres, millimetres, feet, decimal feet or other project-defined units.

Controls:

- identify project units before measurement or import;
- record units explicitly;
- confirm instrument and software settings;
- never assume a CAD/GIS file uses the units you expect;
- document conversions;
- verify conversion factors independently;
- avoid mixing grid, ground and local values without understanding the project method.

Unit errors can create large construction or boundary consequences.

Control points and benchmarks

Control provides the reference framework for measurements.

Before using a control point:

- verify identification;
- verify the coordinate/elevation source;
- confirm datum/reference frame where applicable;
- inspect whether the monument/mark appears disturbed;
- follow project procedures for independent checks;
- record the setup and observations;
- escalate discrepancies rather than silently choosing a value.

Never fabricate or “correct” an authoritative control point because another measurement looks more convenient.

Total stations

A total station combines angle measurement with electronic distance measurement and can calculate/store coordinate information.

A controlled setup may include:

1. inspect equipment, tripod, tribrach, prism/target and batteries;
2. identify the correct occupied point and backsight/reference;
3. centre and level the instrument;
4. enter/check project units, constants, coordinates and atmospheric settings where required;
5. orient the instrument using the approved procedure;
6. take check observations;
7. collect points using required coding/description standards;
8. review unexpected measurements before moving the setup;
9. preserve raw observations and field notes;
10. transfer data using approved storage and naming controls.

Equipment software may automate calculations, but it does not know whether you occupied the wrong control point or entered the wrong prism constant.

Levels and elevation work

Optical, digital or laser levels may be used for elevation transfer and construction control depending on project requirements.

Key controls include:

- stable setup;
- clear sight lines;
- properly held/verified rod;
- known benchmark/reference;
- balanced sight distances where procedures require them;
- correct reading sequence;
- arithmetic checks;
- closure checks;
- documentation of turning points;
- protection against disturbed points.

Do not report elevation precision beyond what the method, equipment and control network can support.

GNSS/GPS

GNSS can provide powerful positioning, but a coordinate is not automatically survey-grade.

Quality can be affected by:

- satellite geometry;
- obstructions;
- multipath;
- correction source;
- antenna height;
- base/rover setup;
- network/service availability;
- datum/reference frame;
- transformation parameters;
- observation duration;
- equipment configuration;
- environmental conditions.

Record the method and quality indicators required by the project. Do not treat a consumer-phone location, autonomous receiver position or unverified RTK result as equivalent to validated professional control.

Coordinate reference systems

Survey and mapping data can become unusable when coordinate-system metadata are missing or wrong.

Understand the difference among:

- geographic coordinates such as latitude/longitude;

- projected coordinates;
- datum/reference frame;
- projection;
- grid zone;
- vertical datum/elevation reference;
- local/project coordinate system;
- transformation/reprojection.

Two datasets can contain reasonable-looking numbers and still be incompatible.

Before merging or comparing data, verify the coordinate reference, units, epoch/date where relevant, transformation method and project specifications.

Traverses and closure

Traverses connect a sequence of measured lines/angles.

A technician should understand conceptually:

- open versus closed traverse;
- angular closure;
- coordinate/linear closure;
- misclosure;
- adjustment methods at the level required by the employer;
- independent checks;
- why a mathematically adjusted traverse can still be wrong if the control or observations are wrong.

Never hide a poor closure by distributing error without documenting the discrepancy and following the project's acceptance criteria.

Construction staking and layout

Construction layout translates design information into field positions.

Before staking:

- confirm current approved drawings/model;
- verify revision number/date;
- confirm coordinate and elevation control;
- understand offsets and tolerances;
- identify conflicts or ambiguous dimensions;
- use independent checks;
- label/record stakes or points consistently;
- document what was set and from which design source;

- escalate discrepancies before construction proceeds.

A technically accurate stake based on an obsolete drawing is still wrong.

Field notes and data lineage

A defensible record should preserve enough information for another qualified person to understand what happened.

Depending on the project, record:

- date/time;
- project/site;
- crew;
- weather/conditions where relevant;
- equipment identifiers;
- setup/control points;
- instrument/target heights;
- calibration/check status where required;
- raw observations;
- point codes/descriptions;
- sketches/photos where authorized;
- unusual conditions;
- corrections/repeated observations;
- file names and transfer paths;
- who reviewed or accepted the work.

Do not overwrite raw observations merely to make results look cleaner.

Quality assurance and quality control

Survey QA/QC can include:

- independent check shots;
- repeated measurements;
- closure checks;
- coordinate/elevation comparisons;
- instrument checks;
- code/description review;
- duplicate-point review;
- outlier investigation;
- drawing/model comparison;
- revision verification;
- file integrity and naming controls;
- peer review;

- documented acceptance criteria.

A software warning is evidence to investigate, not permission to ignore a result.

CAD and civil design software

Current U.S. employer-posting signals for this occupation include:

- Autodesk AutoCAD Civil 3D — **23%**;
- Autodesk AutoCAD — **20%**;
- Bentley MicroStation — **13%**.

Technicians may use CAD/civil software to:

- import survey points;
- create surfaces;
- draft plans/profiles;
- manage point groups/codes;
- prepare contours;
- support alignments/corridors where assigned;
- compare design versus as-built information;
- export deliverables.

Do not alter authoritative design geometry or survey-control values outside your authorization simply because the software allows editing.

GIS and mapping

O*NET posting data also show GIS systems and ESRI ArcGIS among current technology signals.

GIS work may include:

- coordinate-system management;
- layer editing;
- map production;
- attribute data;
- spatial joins;
- quality review;
- integration of survey points with other datasets;
- web-map or database support where assigned.

Survey data and GIS data can have different accuracy and legal-status expectations. The presence of a parcel line in GIS does not necessarily mean it is a legal survey boundary.

Photogrammetry and imagery

Photogrammetry derives measurements or mapped information from photographs/imagery.

Technician work may involve:

- image organization;
- control/check points;
- aerial triangulation support;
- orthophoto/orthomosaic processing;
- feature extraction;
- surface/model generation;
- map compilation;
- QA against independent reference data.

Check acquisition date, resolution, camera/sensor information, control quality, coordinate system and processing settings before interpreting results.

Drones / remotely piloted aircraft

BLS notes that surveying and mapping technicians may use drones to take photographs.

Drone competence and survey competence are separate.

United States

Commercial/Part 107 operations require the appropriate FAA Remote Pilot Certificate and compliance with applicable operating rules.

Canada

Transport Canada uses Basic, Advanced and Level 1 Complex pilot certificates depending on the operation.

Before flight:

- confirm pilot/operator authority;
- confirm airspace and operational category;
- follow registration/marking requirements where applicable;
- inspect aircraft/batteries;
- assess weather and site hazards;
- establish control/check points appropriate to the mapping method;
- protect privacy and restricted locations;
- preserve flight/source metadata;
- validate mapping outputs against suitable reference data.

A drone-derived surface or orthomosaic is not automatically legal survey evidence.

Data security and privacy

Survey and mapping files can reveal:

- private property layouts;
- critical infrastructure;
- utilities;
- construction sites;
- security features;
- client assets;
- personal addresses;
- restricted cultural/environmental resources.

Controls may include:

- approved storage;
- least-privilege access;
- encrypted transfer where required;
- project naming conventions;
- device protection;
- controlled sharing;
- backups;
- retention rules;
- incident reporting;
- removal of sensitive metadata from public deliverables when required.

Do not upload protected coordinates, client drawings or site imagery to personal cloud storage or unapproved AI services.

Field safety

Field surveying can expose workers to traffic, heat, rugged terrain, construction equipment, excavation, electrical/utility hazards, water, weather, wildlife, remote locations and heavy equipment packs.

Follow the employer's hazard assessment and site rules.

Heat

OSHA guidance supports:

- adequate cool water/fluids;
- rest breaks;
- shade or other cool recovery areas;

- training;
- emergency planning;
- monitoring for symptoms;
- gradual acclimatization for new/returning workers.

Traffic and construction areas

Use required high-visibility PPE, traffic-control procedures, spotters, safe parking/vehicle practices and designated access routes. Do not stand or set instruments where the crew cannot be protected from moving traffic or equipment.

Utilities and excavation

Never assume an underground or overhead utility location from a map alone. Follow the jurisdiction/project's locate, clearance and excavation procedures.

Lone/remote work

Follow check-in, communications, weather, first-aid, emergency and travel plans. Do not improvise remote access into unsafe terrain simply to obtain one more measurement.

Equipment care and checks

Typical controls:

- transport equipment in protective cases;
- keep lenses/optics clean using approved methods;
- inspect tripods/clamps;
- charge and label batteries;
- protect devices from moisture/extreme temperatures;
- follow manufacturer calibration/check intervals;
- document damaged equipment;
- do not use equipment beyond known serviceability limits.

A technician should not claim field calibration or repair authority that the manufacturer/employer reserves for qualified service personnel.

Responsible AI

Policy-approved AI can help with:

- explaining formulas;
- drafting field-note or QA templates;
- documenting workflows;
- checking non-sensitive code concepts;

- summarizing public specifications;
- generating synthetic practice datasets;
- brainstorming interview questions.

Controls:

- never send protected coordinates, legal-boundary records, client/site data, proprietary plans, credentials or restricted imagery to unapproved tools;
- verify units, coordinate systems and calculations independently;
- verify generated code before running it;
- check hallucinated functions, regulations, standards and equipment procedures;
- preserve source/measurement lineage;
- never let AI invent monuments, control points, authoritative coordinates, survey-grade accuracy or legal boundaries;
- human/domain validation remains required.

AI output is not evidence that a field measurement, calibration, closure or legal determination was performed correctly.

Accessibility and inclusive work practices

Surveying includes physical field requirements, but many roles combine field and office tasks and some work methods can be adapted.

Helpful practices may include:

- larger text/high-contrast plans;
- screen magnification;
- keyboard-accessible CAD/GIS workflows where supported;
- structured checklists;
- step-by-step field procedures;
- clear point-code dictionaries;
- visual plus audible/haptic equipment feedback where available;
- team-based distribution of field tasks;
- text descriptions or tables accompanying maps/diagrams in training materials;
- predictable file naming and folder structures.

Accommodation depends on the essential duties, terrain, equipment, employer and jurisdiction. Accessibility guidance does not remove genuine safety or professional requirements.

United States — official career benchmark

O*NET-SOC 17-3031.00 — Surveying and Mapping Technicians is the direct occupation.

BLS reports:

- typical entry education: **high school diploma or equivalent**;
- moderate-term on-the-job training;
- median annual wage, May 2024: **\$51,940**;
- median hourly wage: **\$24.97**;
- employment 2024: **59,400**;
- projected employment 2034: **62,100**;
- projected growth: **5%**;
- projected annual openings: about **7,600**.

These are national occupation-level figures, not guaranteed starting pay or a forecast for every local market.

United States — apprenticeship and workforce routes

O*NET links Registered Apprenticeship titles including Surveyor Assistant Instrument, Photogrammetric Technician, Geodetic Computator and Chief of Party.

Readers can use Apprenticeship.gov and American Job Centers/CareerOneStop to investigate live opportunities, WIOA-supported training and local workforce resources. Approved titles or locators do not guarantee a current opening or funding eligibility.

Canada

NOC 22213 — Land survey technologists and technicians is the primary comparison.

Job Bank currently reports national wages:

- low **C\$22.00/hour**;
- median **C\$29.75/hour**;
- high **C\$46.00/hour**.

The national summary says college, apprenticeship training of two or more years, or related supervisory experience may typically be required. Requirements and prospects vary by province/territory.

Readers should verify both occupational requirements and any provincial professional-title or survey-practice rules before accepting work that could be regulated.

Colombia

CUOC 31122 — Técnicos en topografía is a direct comparison.

Functions include topographic surveys, georeferencing, level/as-built/excavation control, operation of distance/angle/elevation instruments, digital terrain models, field measurement, calculations, plans/maps, processing topographic/cadastral/geodetic data, road/urban/hydraulic layout and coordination of field activities.

Important Colombia wage limitation

OCUPACOL displays a historical salary range of **COP \$350,000-\$2,910,053** but explicitly states that the underlying data do **not** have statistical representativeness under its methodology. Do not use this as a reliable national salary benchmark.

Colombia — SENA pathways

Tecnólogo — Levantamientos Topográficos y Georreferenciación

Current SENA Betowa information:

- **3,984 hours;**
- Titulada;
- Tecnólogo;
- presencial;
- skills covering planimetric/altimetric surveying, engineering georeferencing, photogrammetry, special works, road/urban/hydraulic layout, digital tools and a practical stage.

Técnico — Soporte de Topografía y Georreferenciación

Current SENA Betowa information:

- **2,304 hours;**
- Titulada;
- Técnico;
- presencial;
- skills covering construction measurements, design staking/replanteo, digital tools, quantitative reasoning, workplace safety/environmental practice and a practical stage.

Cohorts, seats and locations change. Verify Betowa live before relying on an enrollment date.

Chile

ChileValora provides an active progression:

- **Ayudante de Topografía** — qualification level 1, valid through **December 30, 2028**;
- **Topógrafo(a)** — qualification level 2, valid through **December 30, 2028**.

The profiles cover field safety, equipment, measurement support, total-station/level work, georeferencing and construction layout. Certification scope is not the same as unrestricted legal-surveyor authority.

Peru

SENCICO's **Carrera Profesional Técnica — Topografía y Geodesia** is a **3-year** program.

Current curriculum includes topography, CAD, photogrammetry, cartography, GIS, geodesy, cadastral work, satellite imagery, georeferenced databases and practical workplace experience.

The August 8, 2026 admission exam is already past. Readers should verify the next admission cycle.

PRONABEC's Beca Tec materials include SENCICO Topografía y Geodesia among participating programs/campuses. Verify the current scholarship call, eligibility and covered costs before assuming funding.

Brazil

Brazil's CBO includes family **3123 — Técnicos em geomática**, including:

- **3123-05 Técnico em Agrimensura**;
- **3123-10 Técnico em Geodésia e Cartografia**;
- **3123-20 Topógrafo**.

The technical scope includes geodetic/topohydrographic surveys, field project points, cartographic analysis and cartographic-document production.

Training/occupation classification does not automatically settle professional-registration or reserved-practice questions for every service. Verify Brazilian and local requirements for the work being offered.

Safe portfolio ideas

Use public/open or synthetic data and safe environments. Examples:

- total-station workflow diagram with synthetic observations;
- traverse calculation workbook using invented coordinates;
- CAD surface/profile built from synthetic points;
- GNSS quality checklist;

- open-data GIS map with documented CRS;
- photogrammetry workflow using non-sensitive imagery;
- construction-layout QA checklist;
- field-note template showing lineage and independent checks;
- accessibility-enhanced map with text/table alternative.

Do not publish private parcel evidence, employer/client control networks, protected utility locations, restricted site imagery or proprietary survey files.

Four-week starter plan

Week 1 — measurement and coordinate foundations

Learn units, angles, elevation, bearings/azimuths, coordinates, datums, projections and field-note discipline. Practice calculations using synthetic data.

Week 2 — equipment and field workflow concepts

Study total-station, level and GNSS setup concepts, control checks, staking workflow and safety planning. Use manufacturer/public training material or supervised practice; do not improvise high-consequence field work.

Week 3 — data processing and mapping

Practice importing synthetic/open points into CAD/GIS, creating surfaces or maps, checking CRS/units, validating point descriptions and documenting transformations.

Week 4 — QA, portfolio and interviews

Build one safe project with raw observations, calculation checks, a mapped/drafted output and a short QA report. Prepare interview examples that show how you detect and escalate measurement problems.

Interview preparation

Be ready to explain:

- accuracy versus precision;
- total-station setup checks;
- why a coordinate is not automatically survey-grade;
- control points/benchmarks;
- common causes of GNSS error;
- coordinate systems and units;
- traverse closure concept;
- field-note/data-lineage discipline;

- independent check measurements;
- current drawing/revision control for staking;
- CAD/GIS quality checks;
- technician versus licensed-surveyor authority;
- drone authorization boundaries;
- how you would protect sensitive survey data;
- how you would verify an AI-generated calculation or script.

Employer due diligence

Ask:

- What portion of the role is field versus office?
- Which instruments are used?
- Which CAD/GIS platforms are standard?
- Who establishes and approves project control?
- What tolerances/closure standards apply?
- Who reviews field notes and calculations?
- Are legal-boundary services performed, and who has signing/sealing authority?
- Is drone work required, and who holds pilot/operator responsibility?
- What traffic, heat, remote-work and construction-site controls apply?
- Are travel and overnight field assignments expected?
- What training is provided for equipment and software?
- How are raw data, revisions, backups and client files controlled?

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